

Rape Reporting and Media Sentiment in Relation to #MeToo

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Sexual violence is one of the most pervasive and under-reported crimes in the United States, making it difficult for researchers to accurately assess its prevalence and for policymakers to address survivors' needs. Because reporting rates are shaped not only by the occurrence of assaults but also by social attitudes toward victims, understanding the factors that influence survivors' willingness to come forward is critically important. One factor that may play a role is media coverage. News reporting helps shape public discourse, perceptions of credibility, and societal responses to sexual violence, potentially affecting whether survivors feel supported enough to report their experiences.

The #MeToo movement, which went viral in October 2017 following reporting on Harvey Weinstein's sexual misconduct, catalyzed a public movement regarding sexual violence and harassment in the United States. Within weeks, millions of people had used the hashtag across social media, survivors came forward publicly in unprecedented numbers, and media sources dramatically increased their coverage of sexual assault. Did this movement actually change how the media frames sexual violence, and did it correspond to changes in official rape crime statistics?

This project examines two related subjects: first, trends in reported rape rates across U.S. states from 2014-2019, and second, the sentiment of news headlines mentioning sexual assault or related terms in the two years surrounding the #MeToo movement (October 2016-October 2018). The first component provides a necessary foundation for the second.

This project builds on past scholarly work. One past study concluded that #MeToo had a causal influence on sex-crime reporting in the United States for the first three months after the launch of the movement, spanning October to December of 2017 (1). The DiD analysis in this work is inspired by the analysis completed in that study. Another study conducted a thematic literature review of 41 articles to examine how sexual violence is reported in the media after #MeToo, finding that rape myths, victim-blaming language, biased framing, and other ethical shortcomings remain widespread in the period after, despite increased public attention to sexual violence (2). However, systematic, quantitative analysis of sentiment in sexual assault headlines across the #MeToo inflection point is limited. This project extends that work by analyzing both FBI crime data and MediaCloud headline data to test whether the movement's cultural impact registers in statistical terms.

2. Data

2.1 FBI Uniform Crime Reporting Data. This project used FBI Uniform Crime Reports (UCR) annual files on violent crimes by state from 2010-2019 (3). The FBI expanded its definition of rape in 2013, broadening what crimes counted as rape for the first time in decades. This expansion makes it difficult to compare the years before and after the change. Analysis of pre/post MeToo trends therefore focuses on 2014-2019 to remain within the revised-definition window.

The original data source includes nine types of violent crimes, with two columns per type (one giving the total number of reports, and another giving the rate per 100,000). The rows are split into region, geographic division, and state, with a comparison of the current year to the previous year for each crime. This paper used the rape rate per 100,000 residents, with one observation per US state per year. The FBI's removal of the legacy rape definition column starting in 2017 shifts all subsequent columns left by two positions, requiring year-sensitive column indexing in notebook 04 to correctly extract robbery and aggravated assault rates. After accounting for this shift in parsing, all three non-rape violent crimes (murder, robbery, and aggravated assault) were recoverable as consistent series across the full 2014-2019 window and used as control outcomes in the DiD analysis.

The use of this dataset resulted in several limitations. First, UCR data reflects reported crimes only; rape is among the most underreported violent crimes, with estimates suggesting that only 20-30 percent of rapes are reported to police (4). Second, this data only includes yearly rates, not monthly, limiting this work's ability to analyze the impact of the MeToo movement over time.

2.2 MediaCloud Headline Data. The second piece of this analysis uses indexed news article headlines from thousands of US outlets from MediaCloud (5). Data from 2016-2019 was collected, but filtered to October 2016-October 2018 for the pre/post sentiment analysis. The original unit of analysis from MediaCloud includes individual news articles, each with a title, URL, publication date, and media outlet name. This work uses individual article headlines. After keyword filtering (described in Section 3.1), the working dataset contained 67,997 headlines. For the LLM sentiment comparison, a random sample of 2,500 pre-MeToo and 2,500 post-MeToo headlines was drawn due to the time and environmental constraints of LLM processing.

Applying this dataset also carried some limitations. Headline sentiment alone does not capture the full article tone or framing; important context may not be detectable from a headline alone. VADER, the primary rule-based sentiment tool, was designed primarily for social media text and may misread headlines. The LLM refused to score the most extreme negative headlines, introducing the need for a blanket scoring for the entries that the LLM refused that impacts sentiment.

3. Methods

3.1 Data Cleaning and Preparation.

FBI Data: Each annual .xls file was loaded using `pd.read_excel()`, skipping the first five rows of headers. Columns 0, 1, 2, and 7, were extracted, corresponding to Area, Year, Population, and Rape Count per 100k residents, respectively. Area names were forward-filled across group rows, numeric footnote suffixes were stripped using regex (`str.replace(r'\d+', '', regex=True)`), and rows not matching the target year were dropped. Rows were filtered to the 50 states using an exact-match list; regions like "Mountain" and "New England" were excluded. The rape rate column was converted to numeric, and years to integer. The cleaned annual frames were concatenated with `pd.concat`; a relational join was not needed given that the data had the columns all in the same order, allowing them to be stacked vertically. State name abbreviations were added for the choropleth figure. See cells 6-10 of 01-LoadingDataRapeStats_merged.ipynb for full details. For the DiD analysis, Agentic tools were used to extract murder (column 6), rape revised (column 8), robbery (column 12 pre-2017, column 10 post-2017), and aggravated assault (column 14 pre-2017, column 12 post-2017) rates from each file. Column positions for robbery and aggravated assault were determined dynamically by checking for the presence of a legacy rape header in row 3. See 04-DiDMeToo.ipynb for full details.

MediaCloud data: The data was filtered originally in MediaCloud by applying matches to the terms "#MeToo," "Rape", "Sexual assault" and excluding phrases like "Meghan Trainor" (who had a popular, but unrelated, song called 'Me Too' during this period). The collection of sources was "United States - National." The four annual CSVs from 2016 to 2019 were concatenated and filtered to rows whose title or url field matched at least one of 14 keyword patterns (`\brape\b`, `\bmetoo\b`, `\bharassment\b`, `\bmetoo\b`, etc.) using case-insensitive regex searching in both titles and urls. The results yielded 115,525 matching rows out of 470,225 total. This result was saved as the `filtered_metroo_assault.csv` in the datasets folder. Dates of publication were parsed with `pd.to_datetime(..., errors='coerce')`, NaNs were filled with "", and `str.strip()` was applied. See 02-WordAnalysisMeToo.ipynb and 03-SentimentwithOllama.ipynb for full details.

Table 1. Key Variable Definitions

Variable	Description
Rape_Rate_per_100k	FBI-reported rape rate per 100,000 residents
Area	State where incident was reported
Year	2014-2019
compound	VADER compound sentiment score (−1 to +1) for each headline
llm_score	LLM (llama3.2) sentiment score (−1 to +1); −1 = very negative/victim-blaming, +1 = empowering/survivor-centered
before	October 17, 2016 – October 16, 2017 (the pre-#MeToo period)
after	October 17, 2017 – October 16, 2018 (the post-#MeToo period)
publish_date	Date of article publication
crime_type	One of: rape, murder, robbery, aggravated_assault
is_rape	1 if crime_type == rape, 0 otherwise (DiD treatment indicator)
post	1 if year is greater than or equal to 2018 (DiD post-treatment indicator)

VADER customization: The default VADER lexicon was extended with 14 domain-specific terms (e.g., *survivor*: +1.0; *brave*: +2.0; *alleged*: -1.0; *reckoning*: +1.0) to better capture sexual-assault reporting conventions.

LLM scoring: Each headline was passed to llama3.2:latest via the Ollama local API with a structured prompt instructing the model to return a single number between -1.0 and +1.0. Refusal responses were assigned -1.0, as the model would reject scoring headlines that it deemed to be extreme "victim blaming," or similar negative reasoning.

3.2 Statistical Tests.

Test 1 (reported rape rates): A two-sample *t*-test comparing mean annual national rape rates for 2014–2016 versus 2018–2019.

Test 2 (sentiment shift): A one-sided Mann-Whitney *U* test comparing compound/LLM scores before and after October 17, 2017. Effect size was computed as the rank-biserial correlation: $r = 1 - \frac{2U}{n_1 n_2}$.

Test 3 (DiD- rape vs. other violent crimes): A two-way fixed effects difference-in-differences regression using a stacked long panel of four crime types (rape, murder, robbery, aggravated assault) across 50 states from 2014–2019. The unit of observation is state $\times$ year $\times$ crime type. The estimating equation is: $\text{rate} \sim \text{post} \times \text{is_rape} + C(\text{state_crime}) + C(\text{year})$, where *state_crime* is a state $\times$ crime-type fixed effect absorbing all time-invariant differences between states and between crime types. The coefficient on *post* $\times$ *is_rape* is the DiD estimator: the differential change in rape reporting rates after October 2017 relative to the contemporaneous change in other violent crimes. A second specification restricted to rape versus murder only was estimated as a robustness check; murder rates are not dependent on a person's decision to report, while rape rates are. The parallel trends assumption was tested by fitting an interaction of a linear pre-period time trend with *is_rape* on the 2014–2017 subsample.

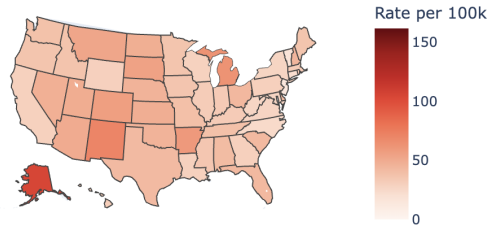
1. Results

Table 2. Analysis sample summary

Source	N (observations)	Years	Unit
FBI UCR (cleaned)	500 state-years	2014–2019	State $\times$ year
MediaCloud headlines (filtered)	~68,000	2016–2019	Headline
VADER pre/post sample	20,249 / 47,748	Oct 2016–Oct 2018	Headline
LLM sample	2,500 / 2,500	Oct 2016–Oct 2018	Headline
DiD long panel from FBI UCR	1,200	2014–2019	State $\times$ crime type $\times$ year

Finding 1: Reported Rape Rates Rose 16% from 2014 to 2019. The mean national rape rate (per 100,000 residents) increased from 36.6 in 2014 to 42.6 in 2019- a 16.4% increase over five years. This rise, specifically with a jump around the time of the MeToo Movement, was statistically significant: a two-sample *t*-test comparing 2014–2016 rates against 2018–2019 rates yielded $p = 0.0178$. The increase was broadly distributed across states rather than driven by a single outlier; Alaska consistently recorded the highest rates throughout the period, while states like New Jersey remained among the lowest, as shown in Figure 1.

Rape Rate per 100k by State (2014)



Rape Rate per 100k by State (2019)

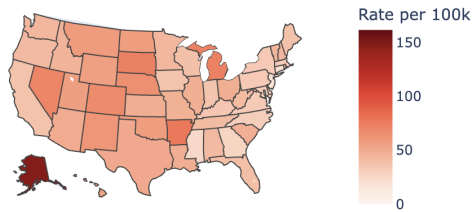


Fig. 1. Top: Reported Rape Rates by State in 2014, ~Three Years Before the #MeToo Movement. Bottom: Reported Rape Rates by State in 2019, ~One Year After the #MeToo Movement.

Rather than reflecting an increase in the prevalence of assault, this rise likely reflects increased reporting—consistent with the hypothesis that the #MeToo movement reduced stigma and encouraged survivors to come forward. This interpretation is consistent with prior research showing that high-profile cases temporarily increase reporting rates, though the sustained multi-year trend observed here suggests a possible, more durable cultural shift. **Importantly, this is a correlational finding; the data cannot determine whether #MeToo caused the increase.**

Finding 2: Rape Reporting Rose Significantly More than Other Violent Crimes After MeToo. The difference-in-differences estimate (the differential post-2017 change in rape rates relative to contemporaneous changes in murder, robbery, and aggravated assault) was +5.71 rapes per 100,000 (95% CI: [+4.30, +7.11], $p < 0.001$) in the full model and +4.15 (95% CI: [+2.66, +5.64], $p < 0.001$) in the murder-only robustness check. The pre-period parallel trends test found no significant divergence between rape and other crime trends from 2014–2017 ($p = 0.29$), supporting the validity of the identifying assumption. The event study (Figure 3) confirms that pre-treatment interactions are near zero and that the differential increase is concentrated in 2018 and 2019. Because all crime types faced the same macro-level reporting environment, state fixed effects, and year shocks, this design isolates the anomalous increase in rape reporting relative to its own counterfactual trend.

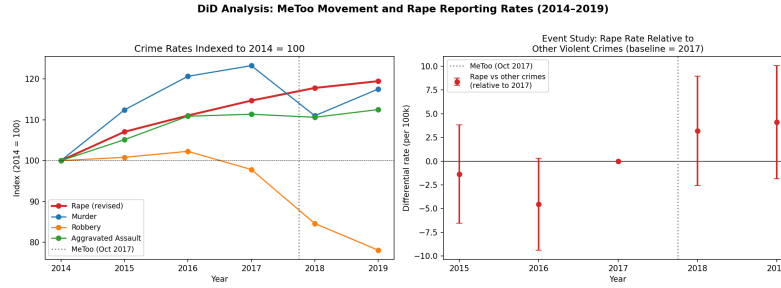


Fig. 2. Left: Mean crime rates for each violent crime type, indexed to 2014 = 100. Right: Event study coefficients for rape relative to other violent crimes, normalized to 2017 as the baseline year. Error bars are 95% confidence intervals.

Finding 3: #MeToo Correlated with a Shift Toward Less Negative Headlines. Both sentiment methods detected a statistically significant positive shift in headline tone after October 2017.

Table 3. Mean sentiment before and after #MeToo, by method

Method	Before	After	Shift	N (before/after)	p-value	r
VADER	-0.69	-0.59	+0.10	20,249 / 47,748	< 0.0001	0.2359
LLM (llama3.2)	-0.80	-0.30	+0.50	2,500 / 2,500	< 0.0001	0.1375

Note: Positive values indicate more empowering/survivor-centered framing; negative values indicate more negative/victim-blaming framing. All coverage remained net-negative throughout the period.

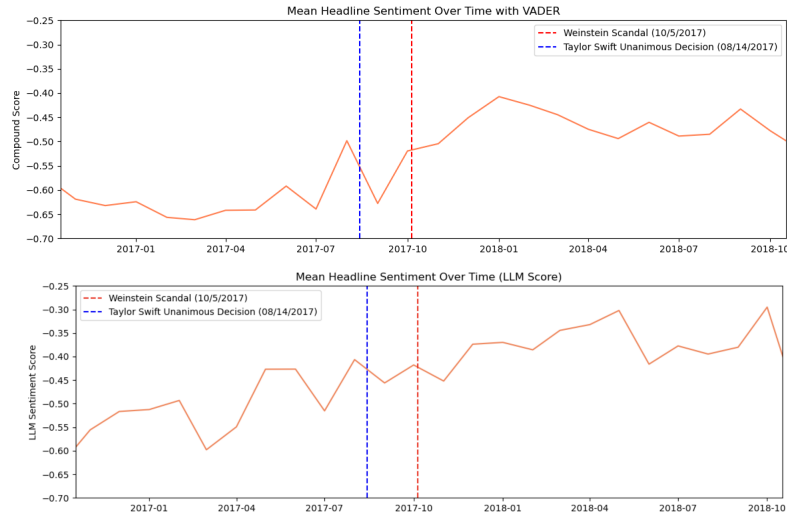


Fig. 3. Top: Mean Headline Sentiment Over Time, Using the VADER Analysis Tool. Bottom: Mean Headline Sentiment Over Time, Using the LLM Analysis Tool.

As stated in Table 3, the VADER method detects a shift of +0.10 (roughly a 14% improvement from baseline), while the LLM method detects a larger +0.50 shift. Both comparisons pass the Mann-Whitney U test at $p < 0.0001$. The VADER tool analyzed a much larger sample of headlines than the LLM did, and effect sizes are small to moderate (VADER: $r = 0.24$; LLM: $r = 0.14$).

The divergence between the two methods is itself informative. VADER, even with domain lexicon customization, tends to score headlines as negative because the subject matter contains inherently negative words. The LLM is better able to distinguish framing; a headline like “Survivor speaks out against abuser” would likely receive a more positive score from the LLM than from VADER. However, the LLM’s refusal to score the most graphically negative headlines (thus coded as -1.0) influences the LLM scores.

Both time series show a clear inflection around October 2017, roughly coinciding with the Weinstein reporting (October 5, 2017). The Taylor Swift groping trial verdict (August 14, 2017) is also marked as a secondary inflection point.

2. Discussion

2.1. Summary of Findings. Taken together, the results paint a consistent picture: the #MeToo movement was associated with a measurable increase in reported rape rates and a statistically significant, if modest, shift toward less-negative sentiment in headlines about sexual assault. The DiD analysis reproduced and extended Levy and Mattsson's findings. Using other violent crimes as the control outcome, a stronger identification strategy than a state-level comparison, the two-way fixed effects estimate found that rape reporting rose approximately +5.7 per 100,000 more than trends in murder, robbery, and aggravated assault would predict. This is slightly different than Levy and Mattsson's estimate, though the methods and sample windows differ. For the media sentiment findings, the compound score shift is statistically significant given the large sample size, but the effect size is small, and the scores remain substantially negative throughout the period. This finding is somewhat unsurprising: news coverage of sexual assault necessarily involves describing violence and trauma, and no cultural moment can make that coverage generically 'positive.' Instead, the upward shift in scores for both models plausibly reflects a change in framing emphasis, which could include more headlines emphasizing accountability or survivor testimony.

2.2. Limitations and Future Directions. Several limitations constrain these conclusions. First, the crime rape reporting data captures only reported offenses and cannot speak to actual prevalence.

Next, the DiD design assumes that rape reporting would have followed the same post-2017 trend as other violent crimes, absent #MeToo- an assumption supported by the parallel pre-trends test but not directly verifiable. One residual concern is that MeToo may have also elevated reporting of aggravated assault, since some domestic violence incidents are classified under both categories; if so, using aggravated assault as a control would cause the DiD to understate the true effect, making the estimate conservative. Murder remains the cleanest control because it is detected independent of victim reporting behavior, and the murder-only robustness check (+4.15 per 100,000) is directionally consistent with the full model. Additionally, because the FBI UCR data are annual, the estimated post period captures 2018 and 2019 as a block rather than isolating the immediate October 2017 effect, which Levy and Mattsson were able to do with monthly data.

Next, the sentiment analysis is limited to headlines, which represent only a fraction of article content and framing. Further, both sentiment methods have systematic flaws: VADER struggles with nuanced journalistic language, while the LLM refuses to score the most negative content, biasing results in opposite directions.

Future work could address these gaps by incorporating full article text in sentiment analysis; attempting LLM scoring with other, more willing models; extending the time window to 2020–2023 to assess whether gains persisted; and using data that would allow for a more long-term analysis before and after the movement.

1. R Levy, M Mattsson, The effects of social movements: Evidence from #metoo. (2023) SSRN Working Paper, last revised May 25, 2025.

2. K Andrews, S Jaffer, S Shariff, Powerful yet disempowered: A thematic literature review exploring the challenges of media reporting on sexual violence. *Trauma, Violence, & Abuse* (2025).

3. Federal Bureau of Investigation, Uniform crime reporting (ucr) program data (2020) Annual state-level violent crime statistics, 2010–2019.

4. RE Morgan, G Kena, Criminal victimization, 2016: Revised, (Bureau of Justice Statistics, U.S. Department of Justice), Bulletin NCJ 252121 (2018).

5. Media Cloud, Media cloud open source media analysis platform (2026).